

A Node SHALL only store a single Client Certificate Details for a particular CCDID. Nodes SHALL provide a mechanism where the CCDID can be used to retrieve, update or remove the CCD associated with the CCDID (see [TLS Client Management Commands](#)).

TLS Client Certificate Signing Request (CSR) Procedure

TLS client certificates can be generated, installed, and rotated using the [ClientCSR](#) and [ClientCSRResponse](#) commands.

The TLS client certificate issuance process uses a freshness check by having the client send a nonce value that must be signed in addition to the signature returned inside the X.509 certificate signing request.

TLS Client CSR Generation, Validation, and Provisioning process

1. The client SHALL generate a random 32 byte nonce using [Crypto_DRBG\(\)](#).
2. The client SHALL send the nonce to the Node and request a CSR using the [ClientCSR](#) command. If rotating, an existing CCDID SHALL be provided.
3. The client SHALL verify the inner signature in the [PKCS #10](#) of the [CSR](#) field in the [ClientCSRResponse](#), per the definition of CSR signatures in [PKCS #10](#).
4. The client SHALL verify [NonceSignature](#) field on the [ClientCSRResponse](#) is valid against the nonce value it provided in the corresponding [ClientCSR](#) command.
5. The client SHALL verify that the public key used for the [NonceSignature](#) is the same public key as the inner signature in the [PKCS #10](#) of the [CSR](#) field.
6. The client performs a procedure not covered in this specification to create and sign a TLS Client Certificate from the returned [CSR](#) field.
7. The client SHALL provision the resulting TLS Client Certificate and any intermediate certificate chain into the NODE using the [ProvisionClientCertificate](#) command using the [CCDID](#) value returned in the [ClientCSRResponse](#).

14.3.2. TLS Client Management

Nodes which have application or utility clusters that use TLS client connections SHALL support the following:

1. [TLS Client Management Cluster](#)
2. [TLS Certificate Management Cluster](#)
3. [Time Synchronization Cluster](#)

This is intended to allow provisioning on the Node of sufficient information to facilitate subsequent TLS client connections and validate the server presented certificates used in those connections.

14.3.2.1. TLS Endpoints

TLS endpoints SHALL be provisioned using the [ProvisionEndpoint](#) command and removed when no longer needed using the [RemoveEndpoint](#) command. In addition, Nodes SHALL provide the following: